CS 173 - Lecture 27 (last one!)

More Countability

- Uncountably infinite: impossible to find bijection $f: \mathbb{N} \to A$ where A is infinite
  - A cannot be put in ordered list indexed by $\mathbb{N}$

ex. Prove $\mathbb{P}(\mathbb{N})$ is uncountable

How to represent every element in $\mathbb{P}(\mathbb{N})$?

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ and $\mathbb{P}(\mathbb{N})$ includes every subset of $\mathbb{N}$

Each element of $\mathbb{P}(\mathbb{N})$ is defined by which elements of $\mathbb{N}$ are in it

$\{1, 2, 3\} \in \mathbb{P}(\mathbb{N}) \to 0111000\dots$ infinitely long bitstring, 0 = not in subset, 1 = in subset

$\{0, 2\} \in \mathbb{P}(\mathbb{N}) \to 101000\dots$

$\emptyset \in \mathbb{P}(\mathbb{N}) \to 000\dots$

Proof by Contradiction: Suppose for contradiction that $\mathbb{P}(\mathbb{N})$ is countably infinite. Then, we should be able to list every element in a list indexed by $\mathbb{N}$

List of all elements of $\mathbb{P}(\mathbb{N})$ represented by infinitely-long bit strings:

	$\mathbb{N}$	$b_0 b_1 b_2 b_3 \dots$
v_0	$0 \to$	$101110\dots$
v_1	$1 \to$	$0100010\dots$
v_2	$2 \to$	$1110110\dots$
v_3	$3 \to$	$100000\dots$
v_4	$4 \to$	$10101011\dots$
	$\vdots$	

new bit-vector: $00010\dots$ (flip bits of diagonal)
differs from every element in the list

Because $\mathbb{P}(\mathbb{N})$ is countably infinite, this list should be comprehensive. However, I can always come up with a new element not in the list by making a new vector which differs from v_k at bit b_k $\forall k \in \mathbb{N}$. We don't have a bijection because we can always find new elements in the co-domain with no pre-image. Because our list of infinitely long bit vectors is not indexable by $\mathbb{N}$, it is not countable, so the set of infinitely long bit vectors is uncountably infinite and thus $\mathbb{P}(\mathbb{N})$ is also uncountably infinite. □

ex. Show $\mathbb{R}$ in $[0, 1)$ is uncountable

$0.\overset{+1}{1}0457\dots$

$0.2\overset{+1}{3}664\dots$

$0.11\overset{+1}{1}111\dots$

$\vdots$

change diagonal to form new number in $\mathbb{R}$ $[0,1)$

$0.\overset{+1}{2}\overset{+1}{4}\overset{+1}{2}\dots$

Things to know about uncountably infinite sets

- If $A \subseteq B$ and A uncountable $\rightarrow$ B uncountable
- If A and B uncountable, NOT necessarily true $|A| = |B|$ (infinite sizes of uncountable sets)
- $|A| < |\mathbb{P}(A)| < |\mathbb{P}(\mathbb{P}(A))|$
  - $\mathbb{N}$ is countable so $|\mathbb{P}(\mathbb{N})|$ has to be larger than countably infinite, so uncountable
  - $\mathbb{R}$ uncountable so $|\mathbb{P}(\mathbb{R})|$ also uncountable

* Review computability from textbook for exam

Discussion Problems – 19.1, Countability Problem

19.1 Which Kind of Infinity?

State whether each of the following sets is countably infinite or uncountable.

(a) $\mathbb{N}$

(b) $\mathbb{P}(\mathbb{N})$

(c) $\mathbb{C}$ (the complex numbers)

(d) $X = \{S : S \subset \mathbb{N}, S \text{ is finite}\}$

(e) The number of books written in Unicode that could ever be written

(f) Real numbers r such that r's decimal expansion eventually ends with the decimal sequence 141592653

a) countably infinite b) uncountable c) uncountable bc $\mathbb{R} \subseteq \mathbb{C}$ and $\mathbb{R}$ uncountable

d) countably infinite e) countably infinite f) countably infinite

Countability additional tutorial problem

We've seen the following two definitions for comparing set cardinalities using **bijections** and **one-to-one functions**:

- Two sets A and B have the same cardinality (|A|=|B|) if and only if there is a bijection from A to B
- |A| ≤ |B| if and only if there is a one-to-one function from A to B.

Do you think there is also a way to compare cardinalities of two sets using **onto functions**? Why or why not? (once you've thought about it, a handwavy justification is fine)

Hint: Given a one-to-one function f:A→B, could we construct an onto function g:B→A? Given an onto g, could we construct a one-to-one f? Hint if you want to formally construct the one-to-one function: Recall that when you have a non-empty set S, you can choose a "representative" element from it (for example we've frequently said things like "let x be an element of S"). You will need to do that here for arbitrarily many sets at once, so you may find it useful to assume that there is a "choice function" *h* which does this for us: *h* takes as input any set and gives as output some element of that set. (The "Axiom of Choice" says that such choice functions do indeed exist.)

If we a one-to-one function $f: A \to B$, then $\exists$ an onto function $g: B \to A$. $|A| \leq |B|$ bc f is one-to-one.